

BS119

2026 Fall Semester
Practice Exercises

Chapter 15

AI may assist you, but it can fail you as well!

□ Use these values where needed

Numerical Values of the Gas Constant R in Various Units

Units	Numerical Value
L·atm/mol·K	0.08206
J/mol·K*	8.314
m ³ ·Pa/mol·K*	8.314
L·torr/mol·K	62.36
L·kPa/mol·K	8.314

* SI unit

□ Use these values where needed

Periodic Table of the Elements

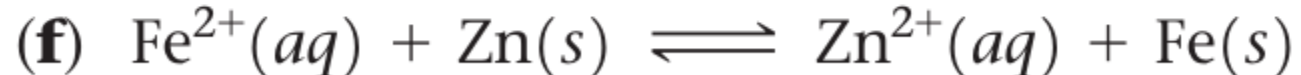
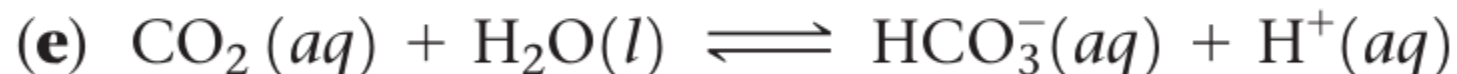
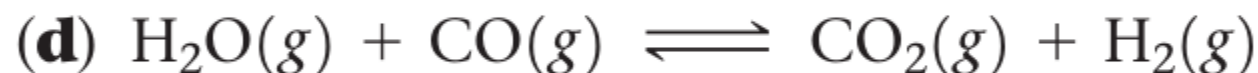
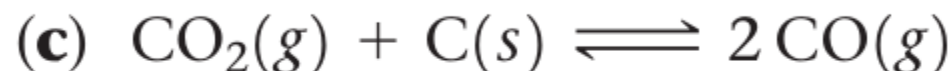
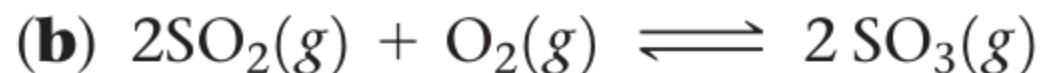
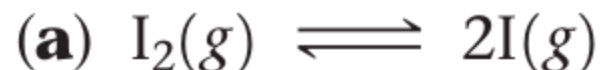
Main Group Representative Elements												Main Group Representative Elements					
1A ^a												8A					
1												18					
1												2					
1.00794												4.002602					
2												10					
6.941												20.1797					
11												13					
22.989770												26.981538					
19												31					
39.0983												69.723					
37												49					
85.4678												114.818					
55												81					
132.90545												204.3833					
87												113					
[223.02]												[284]					
88												114					
[226.03]												[289.2]					
103												115					
[262.11]												[288]					
104												116					
[261.11]												[293]					
105												117					
[262.11]												[294]					
106												118					
[266.12]												[294]					
107																	
[264.12]																	
108																	
[269.13]																	
109																	
[268.14]																	
110																	
[281.15]																	
111																	
[272.15]																	
112																	
[285]																	
113																	
[284]																	
114																	
[289.2]																	
115																	
[288]																	
116																	
[293]																	
117																	
[294]																	
118																	
[294]																	

Lanthanide series	57 La 138.9055	58 Ce 140.116	59 Pr 140.90765	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.964	64 Gd 157.25	65 Tb 158.92534	66 Dy 162.50	67 Ho 164.93032	68 Er 167.259	69 Tm 168.93421	70 Yb 173.04
Actinide series	89 Ac [227.03]	90 Th 232.0381	91 Pa 231.03588	92 U 238.02891	93 Np [237.05]	94 Pu [244.06]	95 Am [243.06]	96 Cm [247.07]	97 Bk [247.07]	98 Cf [251.08]	99 Es [252.08]	100 Fm [257.10]	101 Md [258.10]	102 No [259.10]

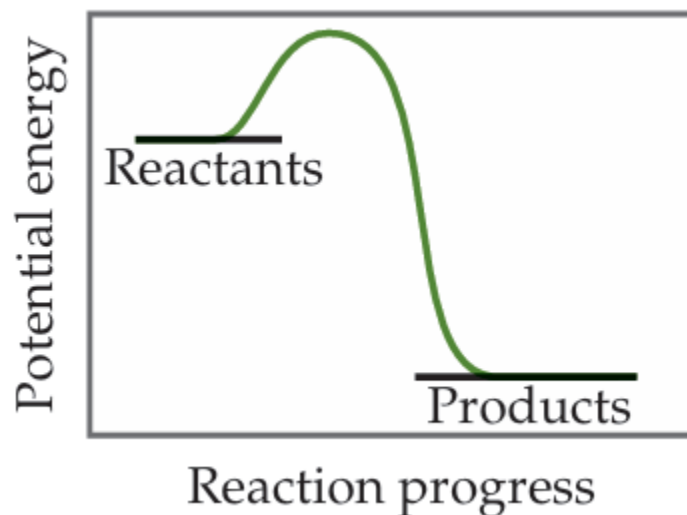
1. Define each of the following terms, phrases, laws, rules, etc. Give at least one example where needed.

- | | | |
|-------------------------------|------------------------------------|-----------------------------|
| ✓ Equilibrium | ✓ Equilibrium position | ✓ Pure solid |
| ✓ Reaction completion | ✓ Homogeneous equilibrium | ✓ Pure liquid |
| ✓ Dynamic equilibrium | ✓ Heterogeneous equilibrium | ✓ Dissolved gaseous species |
| ✓ Static equilibrium | ✓ Concentration change | ✓ Activity |
| ✓ Chemical equilibrium | ✓ Pressure change | ✓ Number of gas molecules |
| ✓ Forward reaction | ✓ Partial pressure | ✓ Total number of particles |
| ✓ Reverse reaction | ✓ Endothermic | ✓ Le Châtelier's principle |
| ✓ Reactants | ✓ Exothermic | ✓ Net reaction |
| ✓ Products | ✓ Enthalpy | |
| ✓ Equilibrium constant, K_c | ✓ Temperature | |
| ✓ K_p | ✓ Equilibrium shift | |
| ✓ Reaction quotient, Q_c | ✓ Catalyst | |
| ✓ Equilibrium expression | ✓ Catalyst effect on reaction rate | |
| ✓ Equilibrium concentration | ✓ Catalyst effect on equilibrium | |
| ✓ Initial concentration | | |

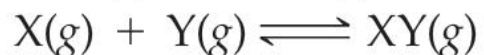
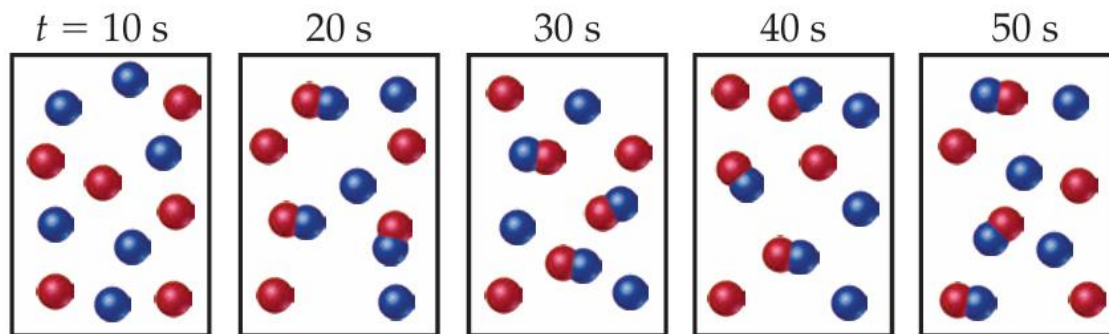
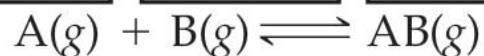
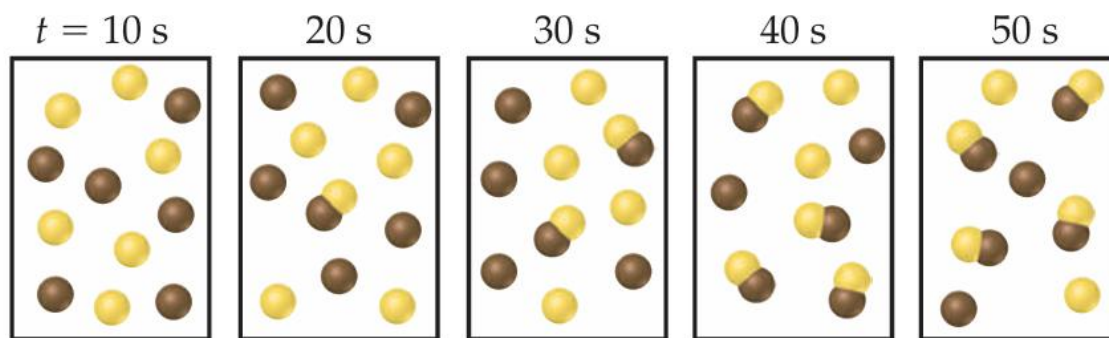
2. Write the expression for K_c for the following reactions. In each case indicate whether the reaction is homogeneous or heterogeneous.



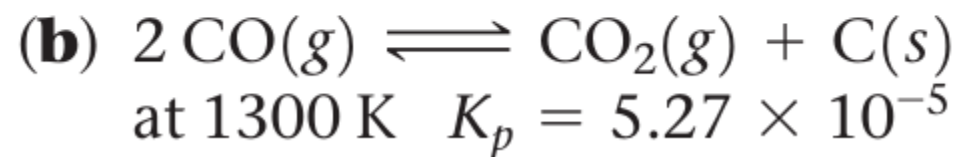
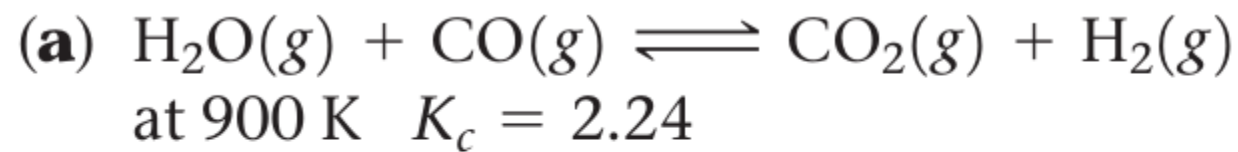
3. Based on the following energy profile for a reversible reaction:
- Predict whether $k_f > k_r$ or $k_f < k_r$.
 - Predict whether the equilibrium constant for the process is greater than 1 or less than 1.



4. Snapshots of two hypothetical reactions, $A(g) + B(g) \rightleftharpoons AB(g)$ and $X(g) + Y(g) \rightleftharpoons XY(g)$ at five different times are shown here. Which reaction has a larger equilibrium constant?

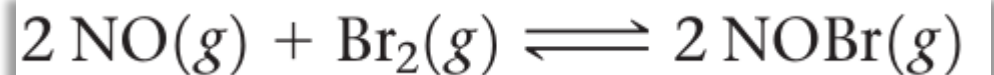


5. When the following reactions come to equilibrium, does the equilibrium mixture contain mostly reactants or mostly products?



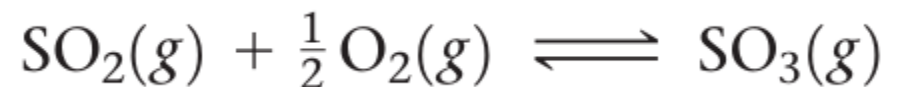
6. When 1.50 mol CO_2 and 1.50 mol H_2 are placed in a 3.00 L container at 395 °C, the following reaction occurs: **$\text{CO}_2(g) + \text{H}_2(g) \rightleftharpoons \text{CO}(g) + \text{H}_2\text{O}(g)$** . If $K_c = 0.802$, what are the concentrations of each substance in the equilibrium mixture?

7. Consider the following reaction at 1000 K.



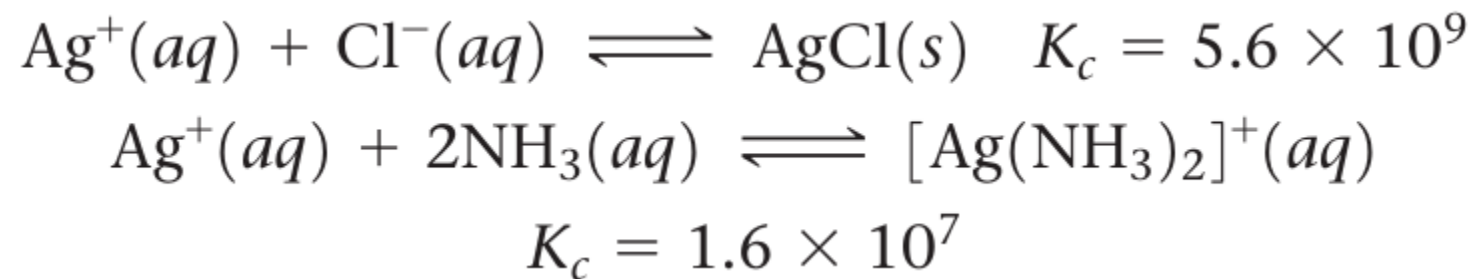
- If $K_c = 1.3 \times 10^{-2}$, what is the value of K_p at the same temperature?
- At this temperature does the equilibrium favor NO and Br_2 , or does it favor NOBr?
- Calculate K_c for: $2 \text{NOBr}(g) \rightleftharpoons 2 \text{NO}(g) + \text{Br}_2(g)$.
- Calculate K_c for: $\text{NOBr}(g) \rightleftharpoons \text{NO}(g) + \frac{1}{2} \text{Br}_2(g)$.

8. Consider the following reaction at 1000 K.

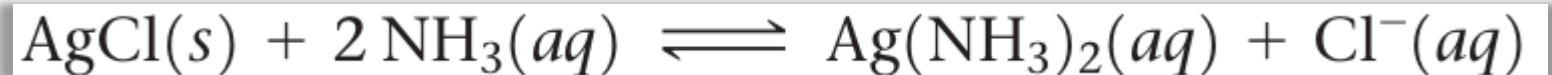


- a. If $K_p = 0.184$, what is the value of K_c at the same temperature?
- b. What is the value of K_p for the reaction: $\text{SO}_3(g) \rightleftharpoons \text{SO}_2(g) + \frac{1}{2} \text{O}_2(g)$?
- c. What is the value of K_p for the reaction: $2 \text{SO}_2(g) + \text{O}_2(g) \rightleftharpoons 2 \text{SO}_3(g)$?
- d. What is the value of K_c for: $2 \text{SO}_2(g) + \text{O}_2(g) \rightleftharpoons 2 \text{SO}_3(g)$?

9. The following equilibria were attained at 298 K.



Based on these equilibria, calculate the equilibrium constant for the following reaction at the same temperature.



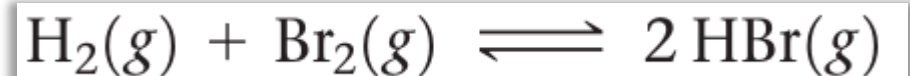
10. Gaseous hydrogen iodide is placed in a closed container at 450 °C, where it partially decomposes to hydrogen and iodine:

$2\text{HI}(g) \rightleftharpoons \text{H}_2(g) + \text{I}_2(g)$. At equilibrium, it is found that $[\text{HI}] = 4.50 \times 10^{-3}$ M, $[\text{H}_2] = 5.75 \times 10^{-4}$ M, and $[\text{I}_2] = 5.75 \times 10^{-4}$ M. What is the value of K_c at this temperature?

11. The equilibrium **$2\text{NO}(g) + \text{Cl}_2(g) \rightleftharpoons 2\text{NOCl}(g)$** is established at 550 K. An equilibrium mixture of the three gases has partial pressures of 10.13 kPa, 20.27 kPa, and 35.46 kPa for NO, Cl₂, and NOCl, respectively.

- Calculate K_p for this reaction at 500 K.
- If the vessel has a volume of 5.00 L, calculate K_c at this temperature.

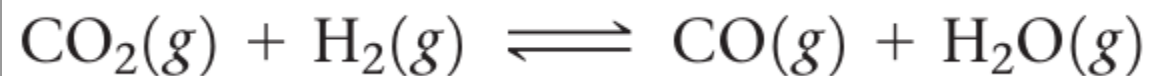
- 12.** A mixture of 1.374 g of H_2 and 70.31 g of Br_2 is heated in a 2.00 L vessel at 700 K. These substances react according to:



At equilibrium, the vessel is found to contain 0.566 g of H_2 .

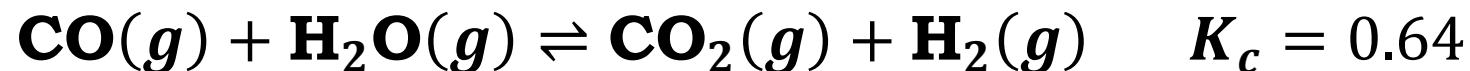
- Calculate the equilibrium concentrations of H_2 , Br_2 , and HBr .
 - Calculate K_c .
- 13.** If $K_c = 1$ for the equilibrium $3\text{A}(\text{g}) \rightleftharpoons \text{B}(\text{g})$, what is the relationship between $[\text{A}]$ and $[\text{B}]$ at equilibrium?

- 14.** A mixture of 0.886 mol of CO_2 , 0.443 mol of H_2 , and 0.713 mol of H_2O is placed in a 10.00 L vessel. The following equilibrium is established at 550 K:



- Calculate the initial partial pressures of CO_2 , H_2 , and H_2O .
- At equilibrium $P_{\text{H}_2} = 182 \text{ kPa}$, calculate the equilibrium partial pressures of CO_2 , CO , and H_2O .
- Calculate K_p for the reaction.
- Calculate K_c for the reaction.

- 15.** Given here are the starting concentrations of reactants and products for three experiments involving the reaction:



Determine in which direction the reaction proceeds as it goes to equilibrium in each of the three experiments shown.

Reactants/Products	Experiment 1	Experiment 2	Experiment 3
$[\text{CO}]_i$	0.0203 M	0.011 M	0.0094 M
$[\text{H}_2\text{O}]_i$	0.0203 M	0.0011 M	0.0025 M
$[\text{CO}_2]_i$	0.0040 M	0.037 M	0.0015 M
$[\text{H}_2]_i$	0.0040 M	0.046 M	0.0076 M

16. Calculate the reaction quotient and determine the direction in which each of the following reactions will proceed to reach equilibrium.

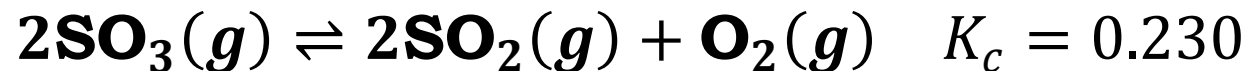
- a.** A 1.00-L flask containing 0.0500 mol of $\text{NO}(g)$, 0.0155 mol of $\text{Cl}_2(g)$, and 0.500 mol of NOCl :



- b.** A 5.0-L flask containing 17 g of NH_3 , 14 g of N_2 , and 12 g of H_2 :



- c.** A 2.00-L flask containing 230 g of $\text{SO}_3(g)$:



17. How do the following changes affect the value of the K_p for a gas-phase endothermic reaction?

- a. Increase in the total pressure by adding a noble gas?
- b. Addition of a reactant?
- c. Increase in the temperature?
- d. Increase in the volume?
- e. Decrease in the temperature?

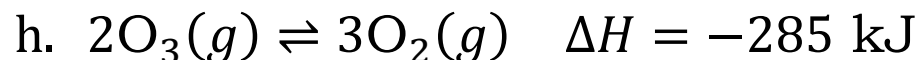
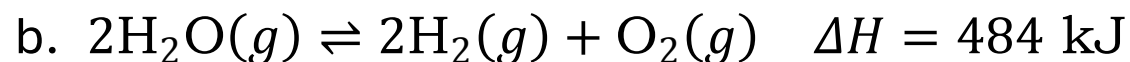
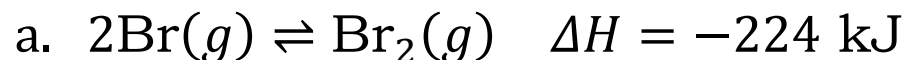
18. Ozone decomposes to molecular oxygen in the stratosphere according to the following reaction.



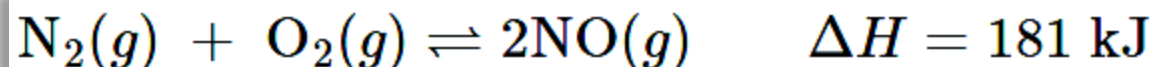
Would an increase in pressure favor the formation of ozone or of oxygen?

19. Consider each of the following reactions.

- i. Explain how an increase in temperature affects each equilibrium.
- ii. Explain how a decrease in the volume of the reaction vessel affects each equilibrium.



20. Nitrogen and oxygen react at high temperatures.



- Write the expression for the equilibrium constant (K_c) for this reversible reaction.
- What will happen to the concentrations of N_2 , O_2 , and NO at equilibrium if more O_2 is added?
- What will happen to the concentrations of N_2 , O_2 , and NO at equilibrium if the temperature of the system is increased?
- What will happen to the concentrations of N_2 , O_2 , and NO at equilibrium if N_2 is removed?
- What will happen to the concentrations of N_2 , O_2 , and NO at equilibrium if NO is added?
- What will happen to the concentrations of N_2 , O_2 , and NO at equilibrium if the pressure on the system is increased by reducing the volume of the reaction vessel?
- What will happen to the concentrations of N_2 , O_2 , and NO at equilibrium if a catalyst is added?